

Roles



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Section Introduction-Roles & Access Control

- What is Access Control Framework and why is it needed ?
- Key Concepts of Access Control Framework.
- Access Control Framework Model
- Types of Roles in Snowflake.
- What is a Custom Role ?
- LAB 1-Create Snowflake Users & Custom Roles.
- What is Role Hierarchy ?
- LAB 2-Create a Role Hierarchy
- Understanding Snowflake Organization and ORGADMIN Role.
- LAB 3-Snowflake Organization
- What is a Database Role ?
- LAB 4-Create & Grant Database Roles.
- What are Secondary Roles ?
- LAB 5-Enable Secondary Roles
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What is Access Control Framework ?

Data security in any database is of highest importance due to the presence of valuable business information stored in them. Cloud databases can be more vulnerable to security incidents like hacking and data theft as access to Snowflake is via a browser which can be accessed from anywhere in the world.

Data Security is broadly categorized into

- Network Security.
- Data Encryption.
- Data Access.

What is Data Access ?

Data Access is the ability to read data from or write data to one or more databases, schema or tables in Snowflake
Data Access is controlled by Access Control framework.



What is Access Control Framework ?

Access Control Framework is a security framework that controls who can access Snowflake data to protect the data from unauthorized access by Snowflake users who are not suppose to see data to which they have not been granted access.

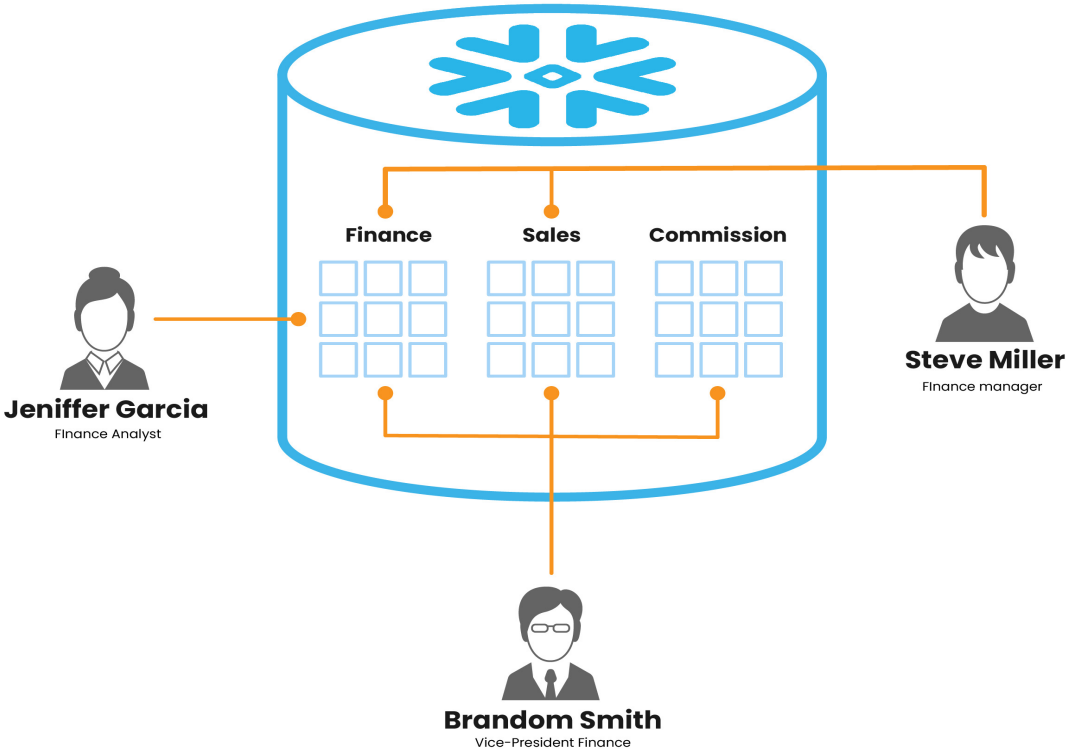
What is Access Control Framework ?

Snowflake User 1 : Vice-President Finance Brandon Smith

Snowflake User 2 : Finance Manager Steve Miller

Snowflake User 3 : Finance Analyst Jeniffer Garcia

	Finance	Sales	Commission
Brandom Smith	✓	✓	✓
Steve Miller	✓	✓	✗
Jeniffer Garcia	✓	✗	✗



Key Concepts in Access Control Framework.

Key concepts in Access Control

- Securable object.
- Role.
- Privilege.
- User.

Securable object.

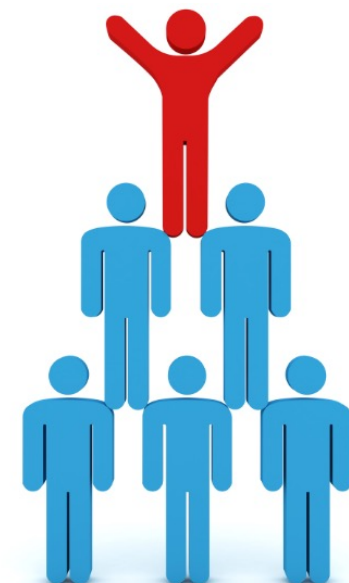
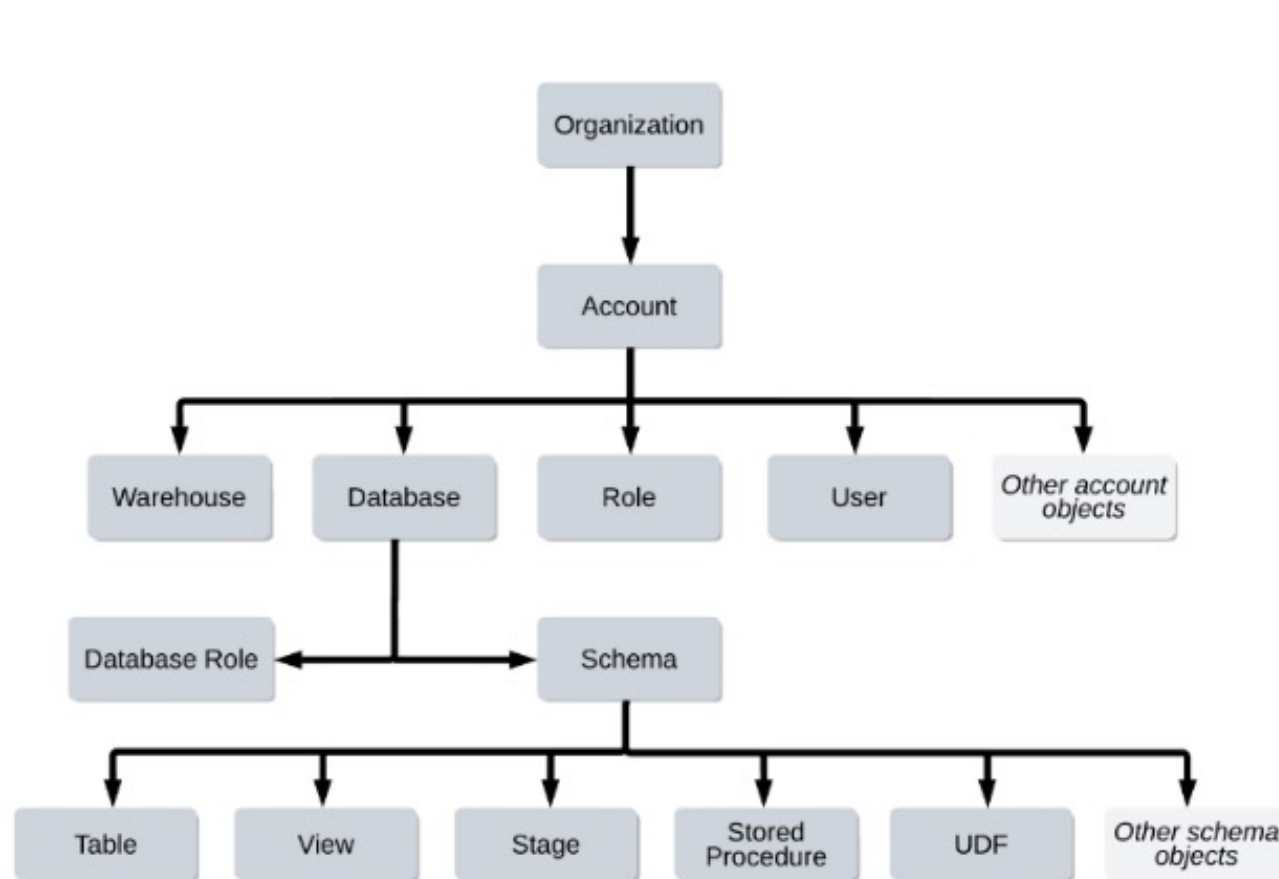
A securable object is a Snowflake object like a table ,database, warehouse or view etc.. which can be accessed only if you have been explicitly given privileges to access the objects. You are not able to access these objects by default just because you are a snowflake user.

Objects are said to be **Securable objects** as they are secured or protected from unauthorized access by default and can only be accessed with the help of Roles.



Key Concepts in Access Control Framework.

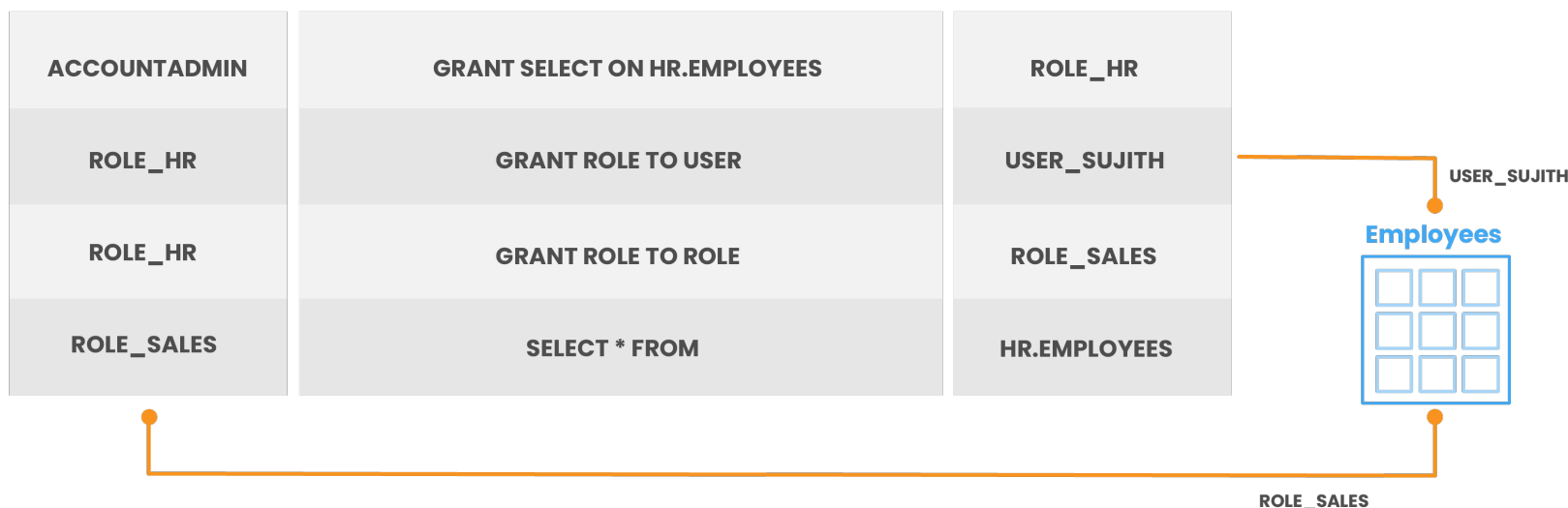
The hierarchy of objects that you see is the list of securable objects in Snowflake and you need explicit access to be given to you so that you can access them.



Key Concepts in Access Control Framework.

Role

Roles are objects to which Privileges on other objects can be given.



- Privileges can be assigned to Roles
- Roles can be assigned to other Users and Privileges in the role are available to the User.
- Roles can be assigned to other Roles and Privileges from the grantor Role are available to grantee Role.

Note : Grantee Role is role that receives the Privilege and grantor is the Role that gives it.

Key Concepts in Access Control Framework.

Privileges

It is the level of access that is provided to an object.

GRANT SELECT ON TABLE

GRANT INSERT ON TABLE

GRANT UPDATE ON TABLE

GRANT DELETE ON TABLE

Employees



TO ROLE PUBLIC

TO ROLE PUBLIC

TO ROLE PUBLIC

TO ROLE PUBLIC



GRANT USAGE ON DATABASE

GRANT USAGE ON SCHEMA



TO ROLE PUBLIC

TO ROLE PUBLIC

Key Concepts in Access Control Framework.

OWNERSHIP

The OWNERSHIP privilege is the highest level of privilege available for a given object and gives the owner full control over the object.

If you assume a role and create an object with that Role the assumed Role will be the owner of the object.



GRANT OWNERSHIP ON TABLE



TO ROLE PUBLIC

CREATE

The CREATE privilege grants the ability to create objects in Snowflake.

CREATE DATABASE
CREATE SCHEMA
CREATE TABLE



TO ROLE PUBLIC
TO ROLE PUBLIC
TO ROLE PUBLIC

Key Concepts in Access Control Framework.

OPERATE WH

- START
- STOP
- SUSPEND
- RESUME

OPERATE TASK

- START
- STOP

GRANT SELECT,INSERT,UPDATE,DELETE ON HR.EMPLOYEES TO ROLE_BI_DEVELOPER



Key Concepts in Access Control Framework.

MODIFY

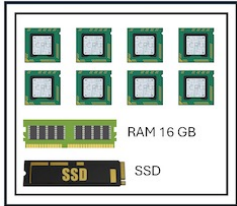
The MODIFY privilege enables alter of tables ,change of Data Types, column Names and addition or removal of columns in tables.

GRANT MODIFY ON TABLE  TO ROLE PUBLIC



OPERATE

Grants the ability to perform operational actions like Starting, Stopping, Suspending and Resuming objects. It applies to Warehouses and Tasks.

GRANT OPERATE ON WAREHOUSE  TO ROLE PUBLIC

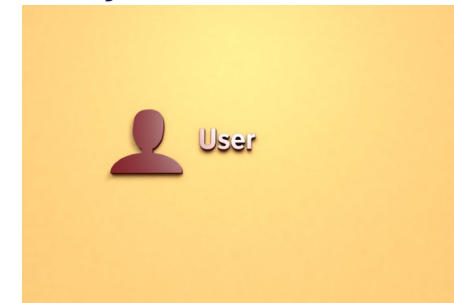
GRANT OPERATE ON  TO ROLE PUBLIC

Key Concepts in Access Control Framework.

User

User is a person or a program that can login to Snowflake and can access securable objects like tables based on the Roles and privileges given to the user.

Roles can be assigned to Users who can assume the Roles assigned to them.



Step 1 Create Users and Roles in Snowflake.

Step 2 Grant privileges like SELECT,INSERT,UPDATE and DELETE on securable objects like DATABASE,SCHEMA,TABLES and WAREHOUSE to Roles.

Step 3 Grant Roles to Users.

Access Control Framework Model

Snowflake's approach to access control combines aspects from both of the following models:

- Discretionary Access Control (DAC)
- Role-Based Access Control (RBAC)

Discretionary Access Control (DAC)

In Discretionary Access Control each object has an owner who in turn can grant access to that object. So if I create a table and I am the owner of the table and I can now grant access to it to any User I like. This approach is called Discretionary Access Control or DAC for Short.

Role-Based Access Control (RBAC)

In Role-Based Access Control access privileges are assigned to Roles, which are in turn assigned to Users. If an object is to be created the User must switch to use a Role with the appropriate privilege to create the object and the Role which the User used to create the object is the owner of the object.

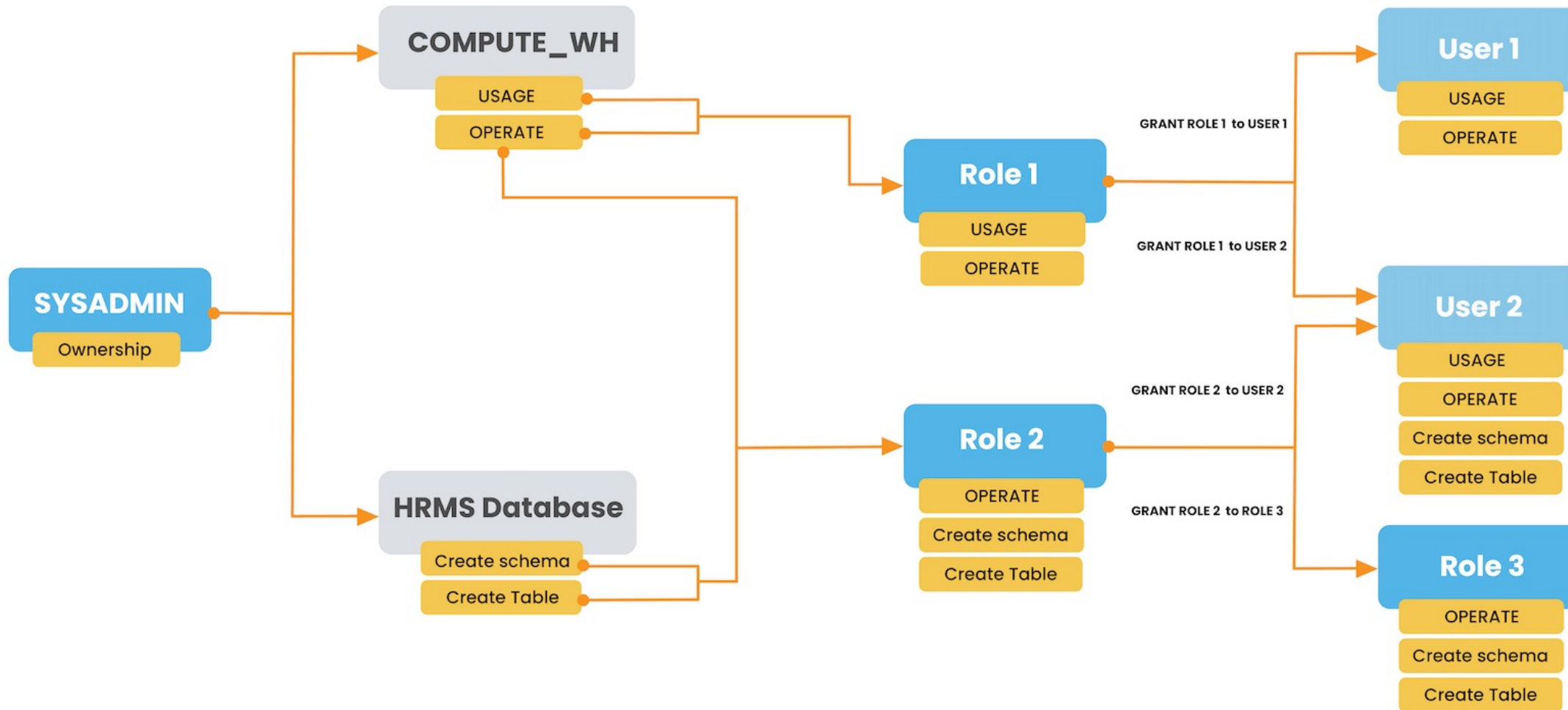
Any User who has been assigned the Role that has ownership of the object will be able to perform all actions that an owner can do when the User has assumed that role.



Access Control Framework Model

Understand RBAC with the help of an example.

- USAGE = Can run Queries
- OPERATE = Start, stop or suspend a WH



Types of Role

Roles are broadly classified into 2 types:

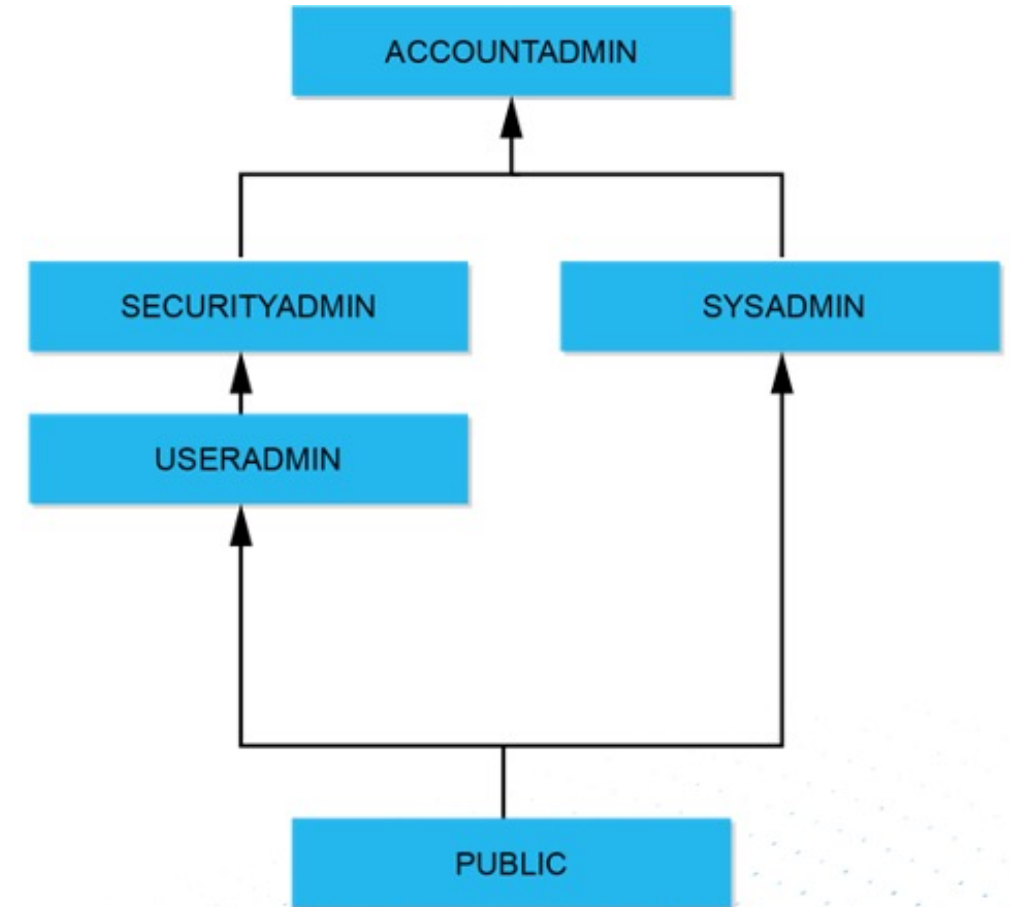
- System-Defined Roles.
- Custom Roles.

System-Defined Roles.

System defined Roles are Roles that are available in Snowflake right of the box and have a specific functionality they are designed to meet.

System-Defined Roles available in Snowflake are

- ACCOUNTADMIN.
- SECURITYADMIN.
- SYSADMIN.
- USERADMIN.
- PUBLIC.



Types of Role

ACCOUNTADMIN:

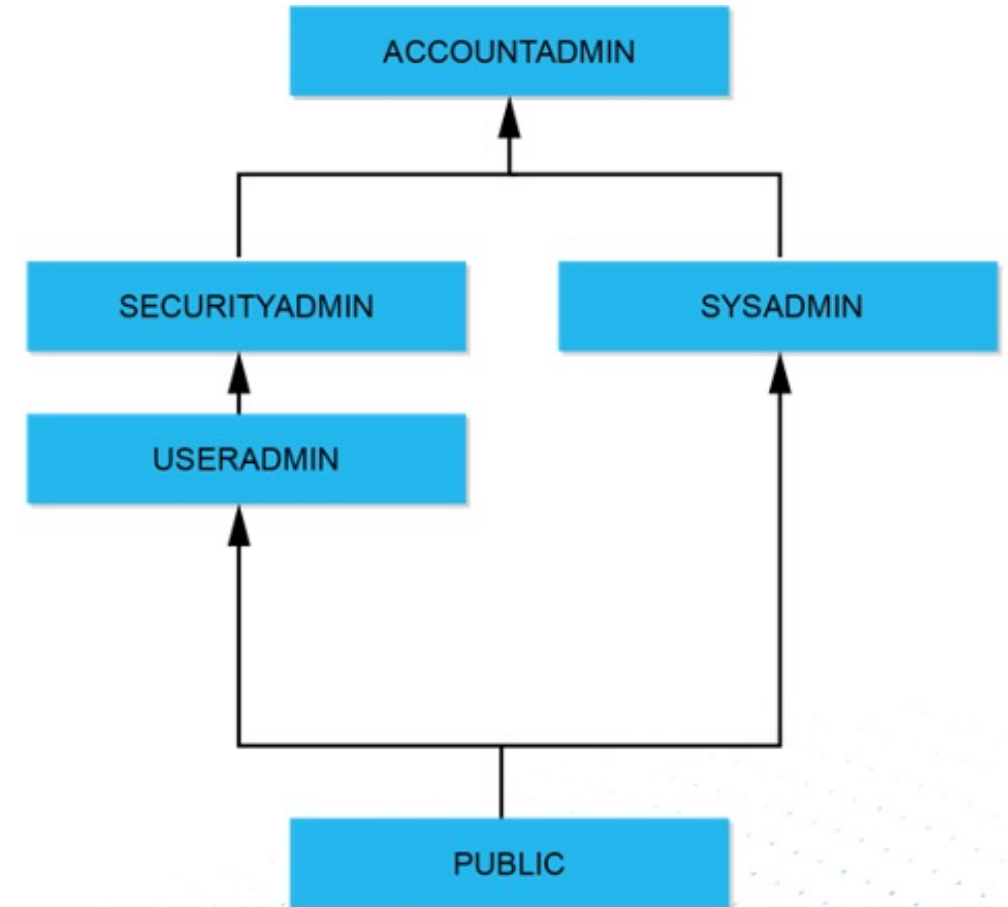
- It is the account administrator and top Role which has access and privilege to all the objects in the account.
- Role should be granted to a small number of users of no more than 3.
- Accountadmin has access to the billing information which is at the account level.

SECURITYADMIN:

- SECURITYADMIN can create Users, Roles , grant privileges on objects that it **does not own** and has the “Manage Grant” privilege.
- SECURITYADMIN manages Network security for Snowflake account and is used to create Network policy

USERADMIN:

- USERADMIN is a Role that can create users and Roles.
- It can manage the user and Roles it creates.



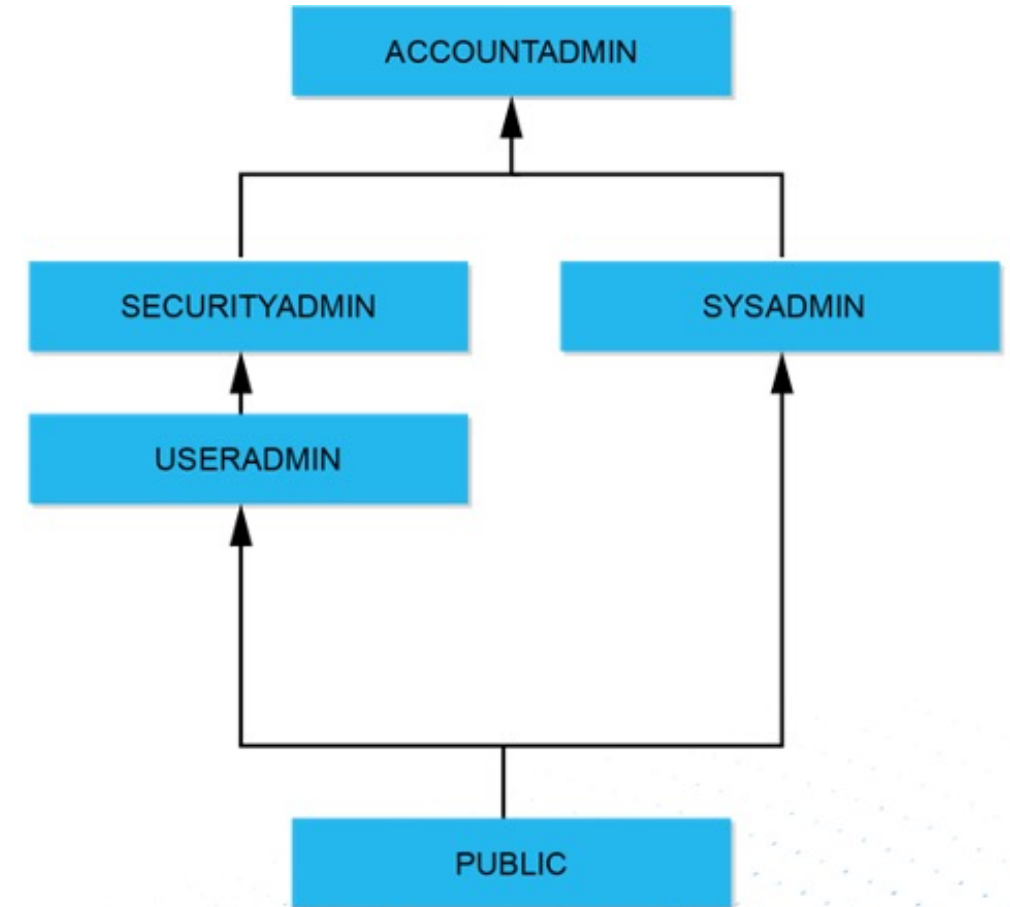
Types of Role

SYADMIN:

- SYADMIN is a role that can create objects like database, schema, tables and warehouses to name a few.

PUBLIC:

- PUBLIC role is automatically granted to every user and every Role in your account.
- If you grant access of an object to the PUBLIC Role all users in your database will have access to the object.

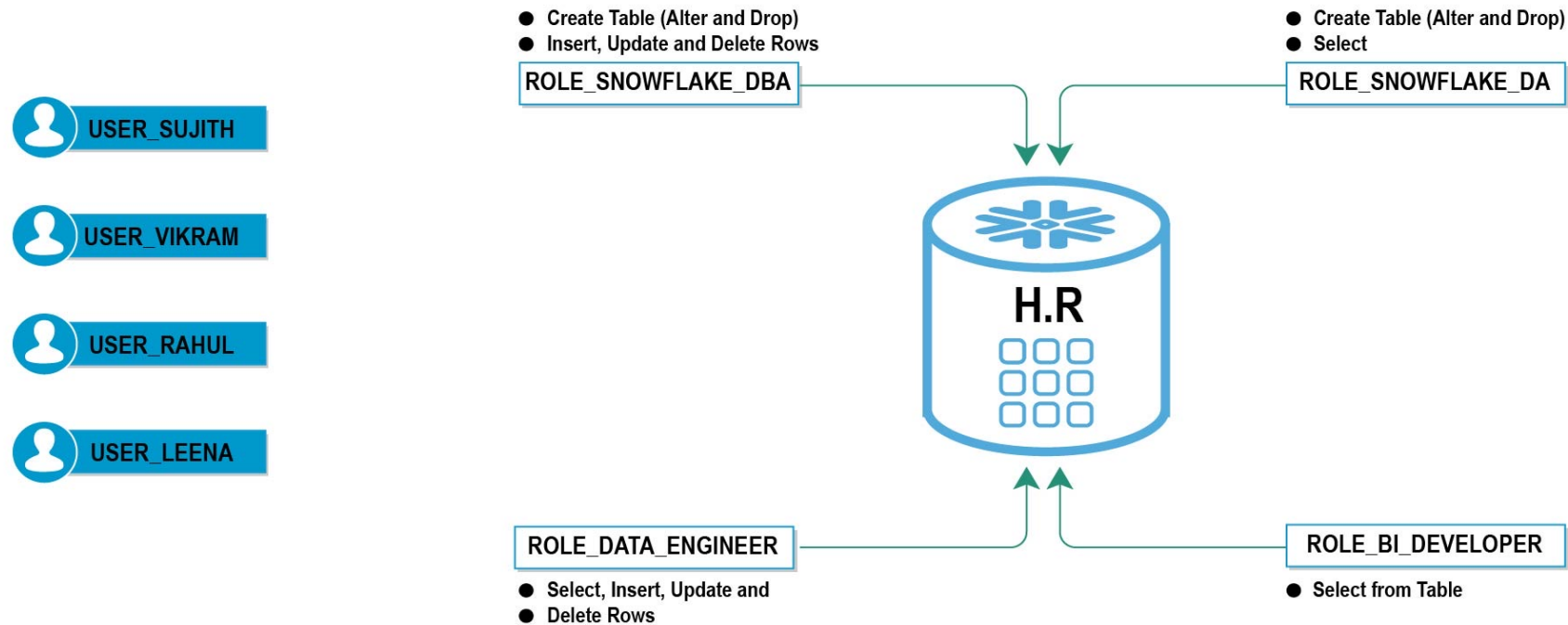


What is a Custom Role ?

A custom role is Role which is created using the CREATE OR REPLACE ROLE command.

Eg: `CREATE OR REPLACE ROLE ROLE_BI_DEVELOPER`

Custom Roles are needed to provide differing level of privileges to different users depending on their job responsibilities in the project.

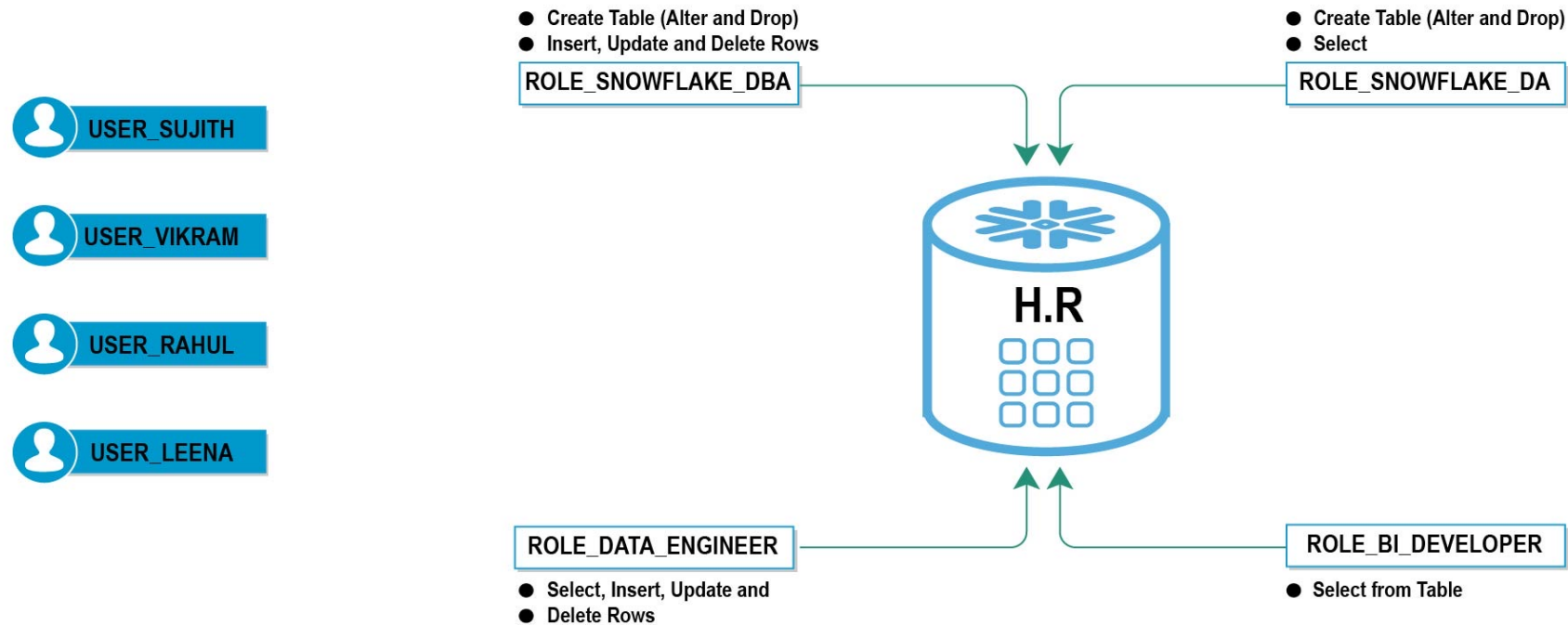


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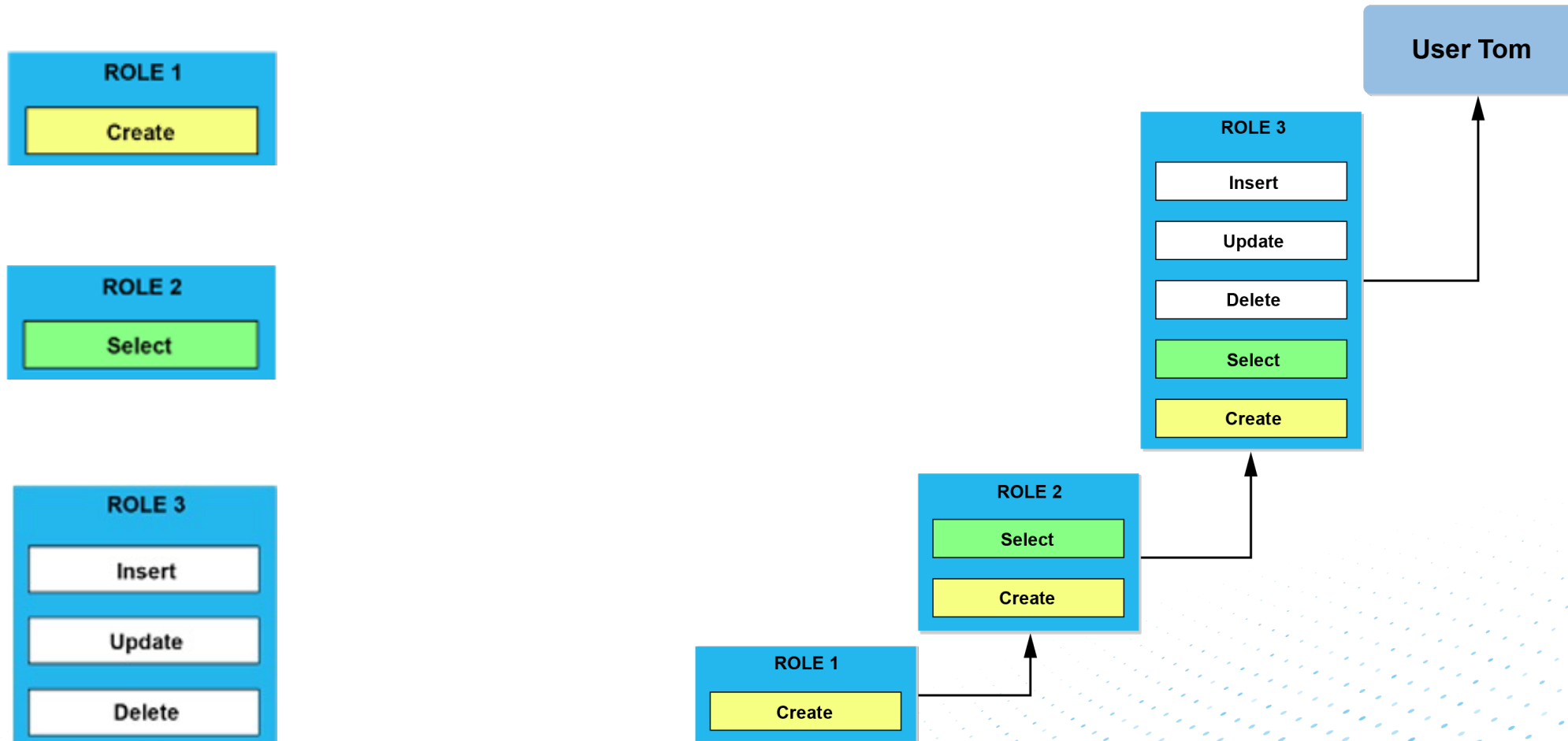
Eg: `CREATE OR REPLACE ROLE ROLE_BI_DEVELOPER`

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What is Role Hierarchy ?

We have 3 roles Role 1, Role 2 and Role 3



What is Role Hierarchy ?

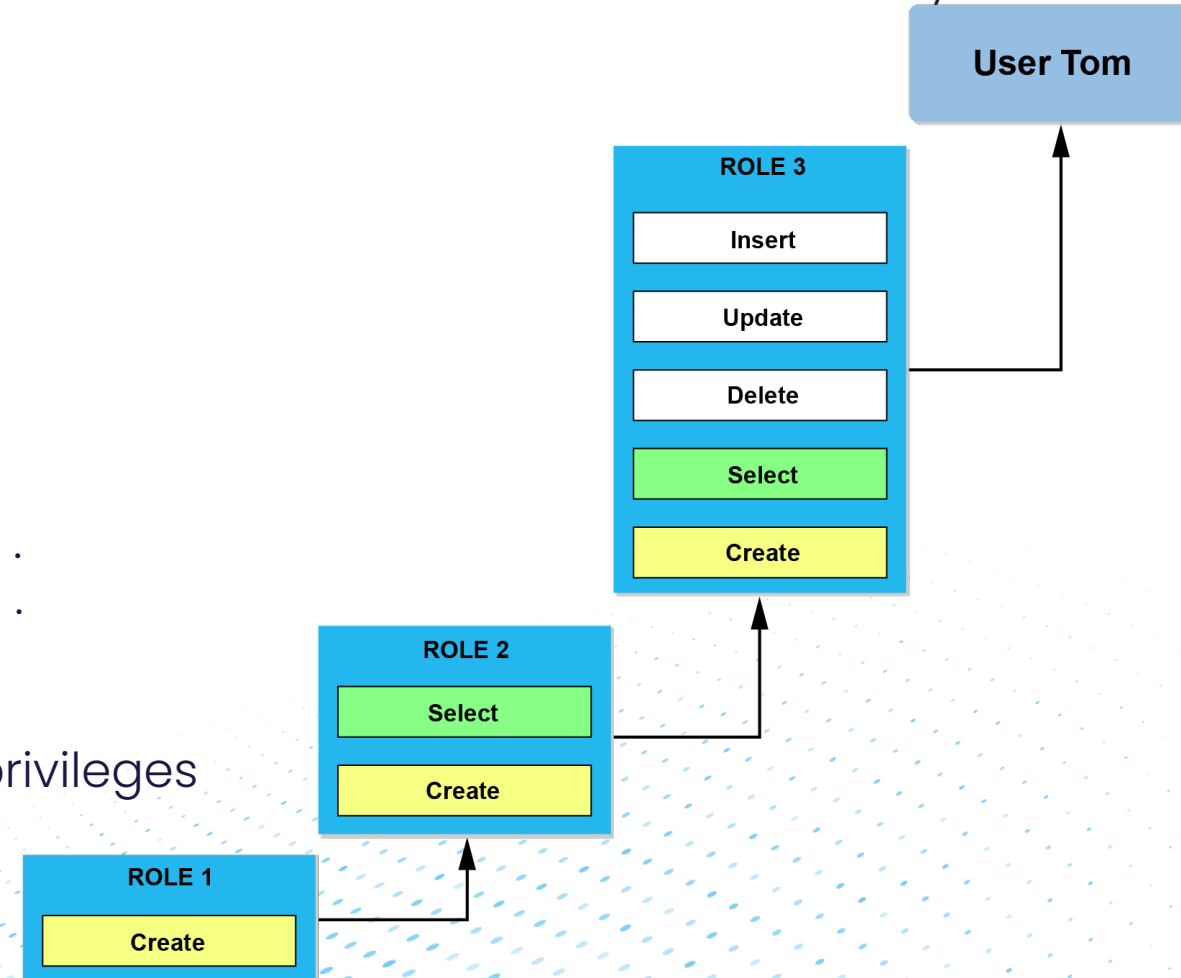
Roles can be granted privileges like SELECT,INSERT,UPDATE etc. and these Roles in turn can be granted to other Roles creating a relationship between them. This process of granting one Role to the other creates a parent child relationship between them and is called a Role Hierarchy.

Role 3 is parent of Role 2
Role 2 is parent of Role 1

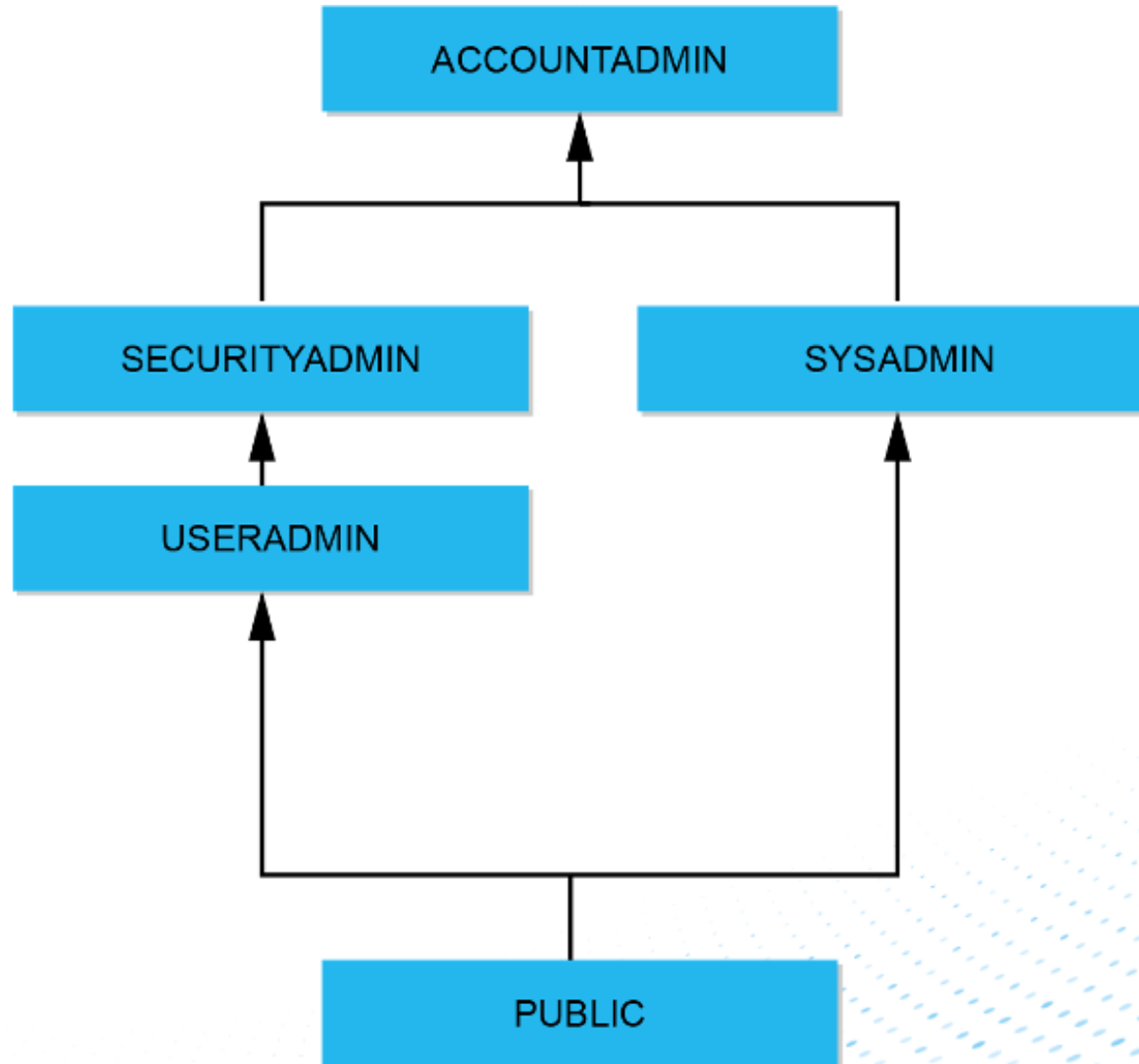
Role 2 is child of Role 3
Role 1 is child of Role 2

All the privileges from Role 1 are inherited by Role 2 .
All the privileges from Role 2 are inherited by Role 3 .

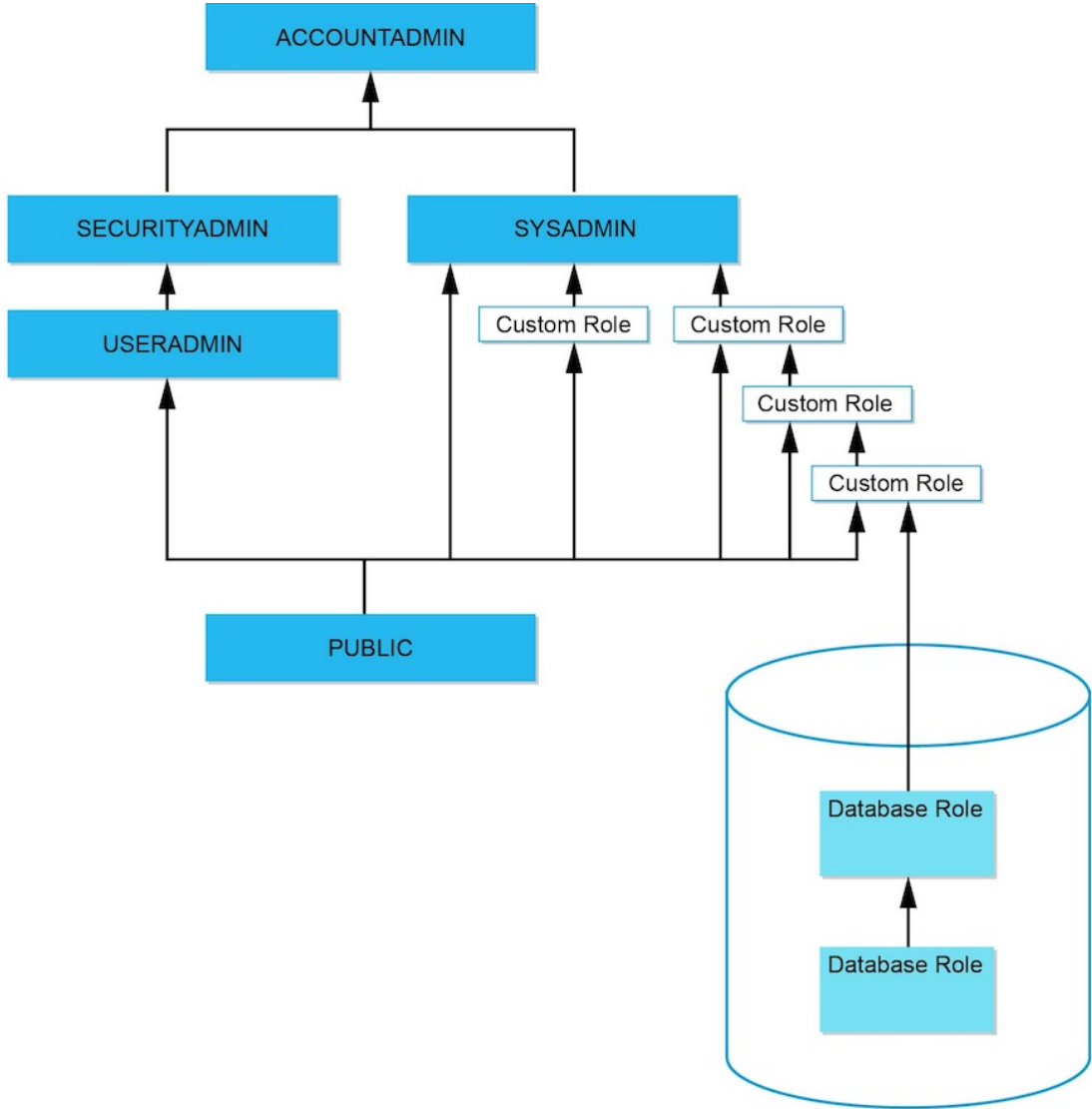
When user Tom assumes the Role 3 he has all the privileges from Role 3 available to him.



What is Role Hierarchy ?



What is Role Hierarchy ?



Snowflake ORGANIZATIONS & ORGADMIN Role?

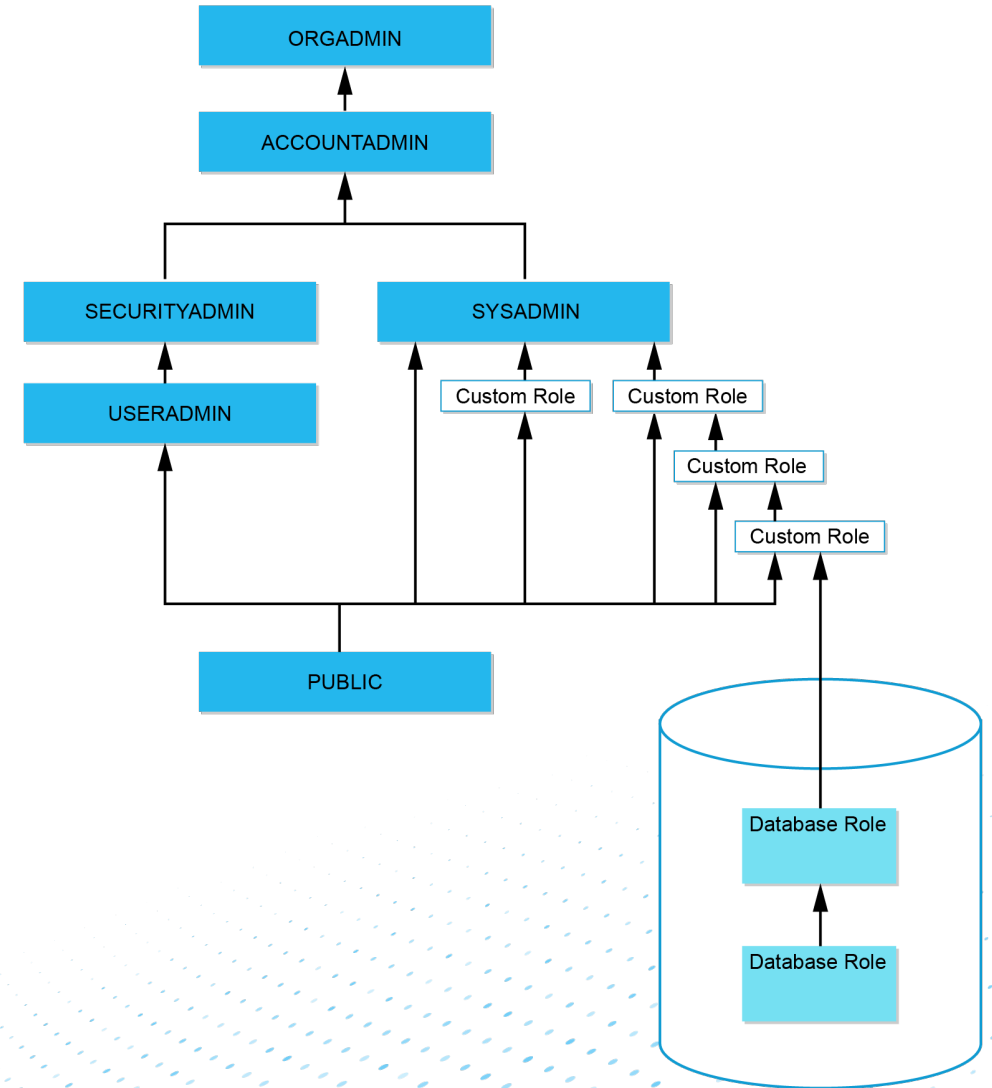
An Organization is a Snowflake object that links accounts owned by a single business entity.

Organizations simplify account management, billing, Snowflake Secure Data sharing, replication and failover and other account administration tasks.

The ORGADMIN Role is created to manage Organizations and can view, create, and manage all your accounts across different countries, regions and cloud platforms.

The ORGADMIN can

- Create new accounts and delete existing ones
- View all accounts in the Organization.
- View all the regions enabled for the Organization.
- View usage information for all accounts.
- Enable replication for an account in an Organization.

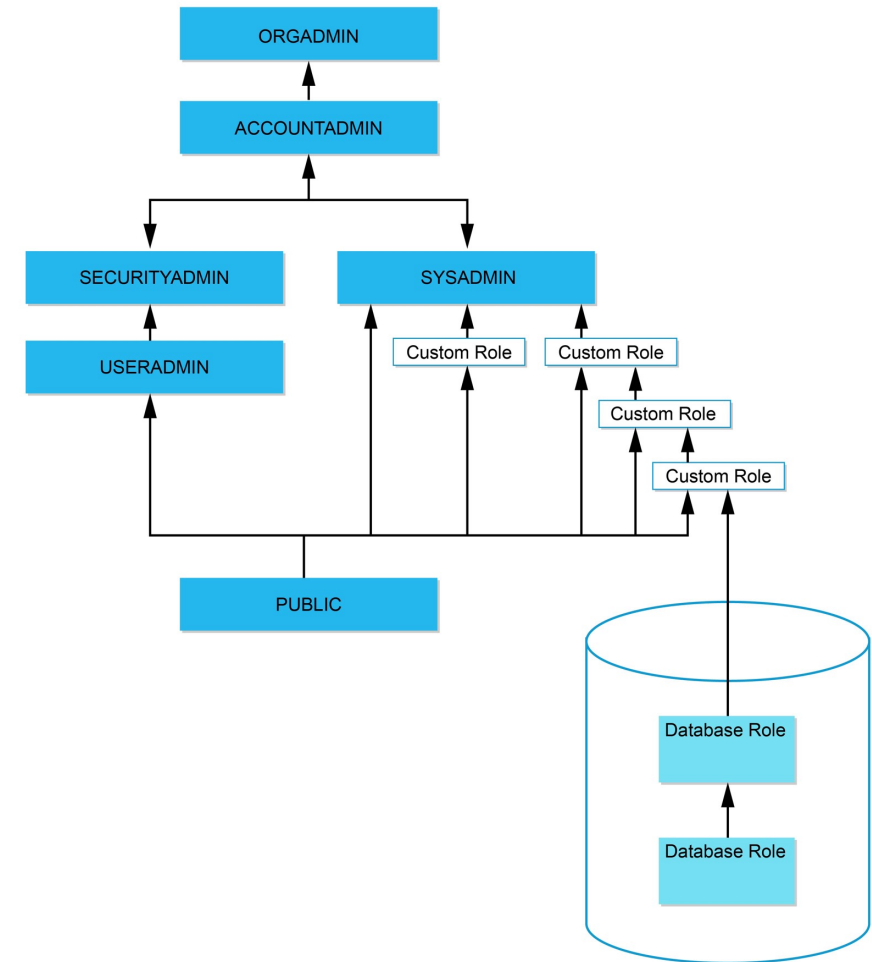


Snowflake ORGANIZATIONS & ORGADMIN Role?

The commands used by ORGADMIN are

- CREATE ACCOUNT.
- DROP ACCOUNT.
- SHOW ORGANIZATION ACCOUNTS.
- UNDROP ACCOUNT.

Snowflake provides historical usage data for all accounts in your Organization via views in the **ORGANIZATION_USAGE** schema in a shared database SNOWFLAKE.



What is a Database Role?

Database Roles help get access to ACCOUNT_USAGE schema in the SNOWFLAKE database which is automatically created when we create a Snowflake Account.

The SNOWFLAKE database stores information about all objects like tables, views and roles and access to this information is available to certain Roles like the ACCOUNTADMIN .

Database Roles are a way to give Data Engineers access to the ACCOUNT_USAGE schema without giving them the ACCOUNTADMIN Role.

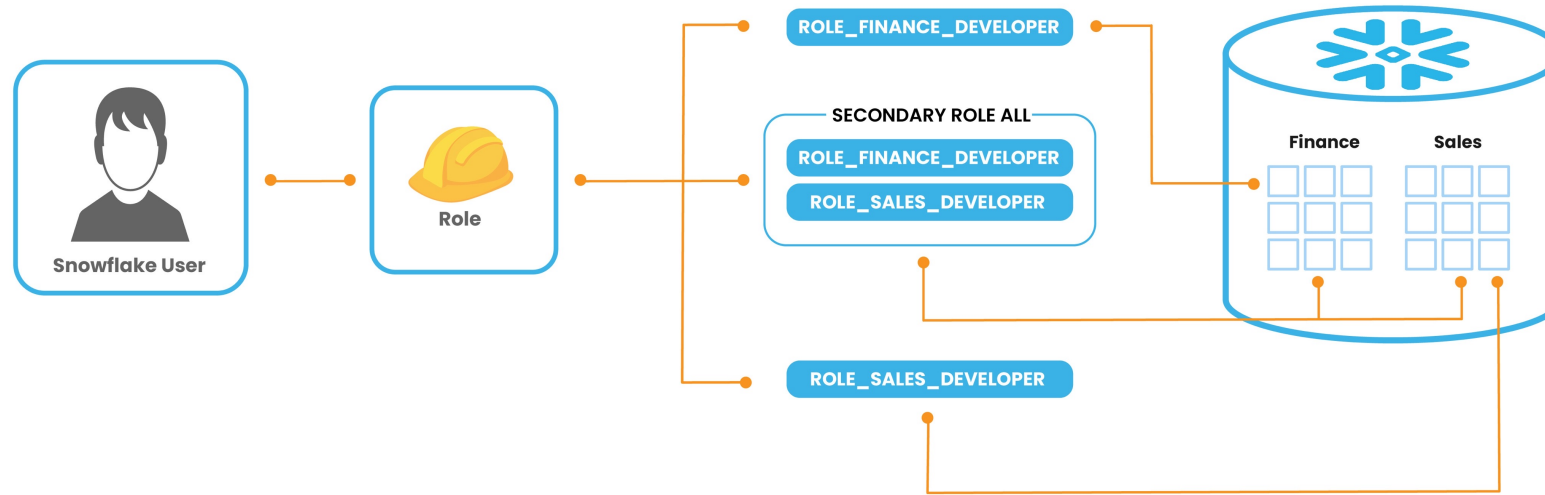
There are several Database roles that are provided which we can grant to data engineers.



What is a Database Role?

DATABASE ROLE	ROLE DESCRIPTION
OBJECT_VIEWER	The OBJECT_VIEWER role provides visibility into object metadata.
USAGE VIEWER	The USAGE_VIEWER role provides visibility into historical usage information.
GOVERNANCE_VIEWER	The GOVERNANCE_VIEWER role provides visibility into policy related information.
SECURITY_VIEWER	The SECURITY_VIEWER role provides visibility into security based information.

What are Secondary Roles ?



Secondary roles are Snowflakes way of allowing a user to use privileges from more than one Role in a given Snowsight session.

Privileges from all Roles can be activated by using `USE SECONDARY ROLES ALL` in Snowsight.

The Role you assume when the Snowsight session starts is called the primary Role and the other Roles which have been assigned to you are called secondary Roles.

Summary

- Access Control Framework are the tools that Snowflake provides to protect your data from unauthorized access from Snowflake users.
- Snowflake supports 2 types of Access Control Framework model Discretionary Access control (DAC) or Role Based Access Control(RBAC).
- A securable object in Snowflake is an object to which access can be controlled with the use of Roles and grants.
- Roles are objects to which privileges on other objects can be given. Roles can be granted to a user or another Role.
- Privilege is the level of access that is provided to an object.
- Roles are broadly classified into SYSTEM-Defined Roles and Custom Roles.
- SYSTEM-Defined Roles are predefined and come out of the box with Snowflake. Custom roles need to be created by a User in Snowflake and appropriate privileges needs to be given to it after which they can be granted to a User.
- System-Defined roles available in Snowflake are ACCOUNTADMIN,SECURITYADMIN,SYSADMIN USERADMIN and PUBLIC.
- Granting a Role to other Role results in the privilege of the grantor role being passed to the grantee role, this creates a parent child relationship between them and creates a Role Hierarchy.
- Role hierarchy exists between SYSTEM-Defined roles with the ACCOUNTADMIN at the top and the PUBLIC role at the bottom of the hierarchy.

Summary

- An **ORGANIZATION** is a Snowflake object that links accounts owned by a single business entity.
- The ORGADMIN is the Role which is able to view and manage all your accounts across different countries, regions and cloud platforms and manage the Organization.
- The ORGADMIN can create new accounts and delete existing ones.
- The ORGADMIN can access the **ORGANIZATION_USAGE** schema to see usage data across the entire organization.
- Database Roles help in getting access to **ACCOUNT_USAGE** schema in the SNOWFLAKE database, typically only the ACCOUNTADMIN Role has access to this schema and granting Database Roles to a User can provide access to the **ACCOUNT_USAGE** without granting ACCOUNTADMIN Role.
- Multiple Database Roles are available to grant to a User, to enable access to credit and storage consumption we can grant the **USAGE_VIEWER** Database Role.
- Secondary roles are Snowflakes way of allowing a User who has been granted multiple Roles to use privileges from more than one role in a given Snowsight session.
- The Role that the User assumes is the primary role and the other Roles to which the User has access are the Secondary Roles.

Summary

GRANT SELECT,INSERT,UPDATE,DELETE ON HR.EMPLOYEES TO ROLE ROLE_DATA_ENGINEER

Thank you



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